Data Integration and Warehousing

Final Project

(Airport Management System)

Projet Scenario:

This project focuses on airport operations and flight management. The selected data sources are related to flight schedules, delays, and cancellations. The purpose of this lab is to identify suitable data sources and perform an initial data profiling to understand the structure and quality of the data before any future integration.

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Project Scenario and Data Sources

This project focuses on airport operations and flight management. The goal is to integrate airline-related datasets and build a data warehouse for analyzing flights, airports, passenger traffic, and airline performance

Source Name	Type	Format	Records	Notes
Flight Delay Data	Structured	CSV	171.667	Contains airline delay information for performance analysis.
2015 Flight Delays and cancellations	Structured	CSV	Airport file 323	Includes airports-level information for US airports.
Air Traffic Passengers Trends (USA)|1999-2024	Structured	CSV	38.197	Shows passenger traffic trends across US airports over time.

- **Flight Delay Data Profiling:**

Flight Delay Data contains 171,667 records with no nulls, but high duplication across most columns due to repetitive monthly reporting

- **Flight Delays Profiling 2015:**

The 2015 Flight Delays dataset has 323 airport records. IATA_code has no duplicates and serves as a unique identifier, while latitude has a 0.93% null rate

- **Air Traffic Trends Profiling:**

The dataset contains 38,197 records and provides passenger traffic information across US airports. Most attributes have no missing values, making the dataset suitable for integration and analysis

Structured vs. Unstructured Data

Unstructured Data Analysis Table

Attribute	Data Type	Structured / Unstructured	Justification
Review_ID	Integer	Structured	It's uniquely identifies each review in numeric format
Airline_Name	Varchar	Structured	It contains predefined airline names
Overall_Rating	Integer	Structured	It is a numeric rating between 1 and 5
Review_Date	Date	Structured	It follows a standard date format
Review_Title	Text	Unstructured	It is short free-form text written by customers
Review	Text	Unstructured	It is detailed free-form customer feedback without a fixed structure

- **Transformation Plan Table:**

Five transformation steps were applied: removing punctuation, lowercasing text, removing stop words, standardizing date format, and creating a Clean_Review column

- **Proposed Structured Output Table:**

The final structured output has six fields: Review_ID (PK), Airline_Name, Overall_Rating, Review_Date, the original Review text, and a derived Clean_Review column

Data Cleaning & Preprocessing

Cleaning Strategy Table

Issue	Cleaning Technique	Rule Applied	Justification	Expected Outcome
Duplicate values in Carrier_Name	Remove duplicates	Drop duplicate carrier names	Duplicate values do not add new information	Reduced redundancy
Inconsistent formatting in Airport_Name	Standardize text format	Clean airport name formating	Improves data usability	Easier geographic analysis
Inconsistent values in Operating Airline	Normalize airline names	Convert to unified naming format (Title Case)	Avoid same airline appearing in different formats	Consistent airline categories
Duplicate values in Passenger Count	Aggregate or keep records	Keep values but ensure correct aggregation during analysis	Passenger count values vary widely across records	Accurate statistical analysis
Duplicate values in Airline Name	Normalize text values	Standardize airline naming and trim spaces	Prevent duplicate airline names with small differences	Consistent airline names
Invalid values in Overall_Rating	Validate rating values	Replace invalid value ('n') with NULL then convert column to numeric	Ensures ratings are valid numbers	Reliable rating analysis

Data Quality Assessment Table:

- Six data quality issues were identified across the datasets.
- Redundancy was found in Carrier_Name, Operating Airline, and Airline Name
- Airport_Name contained formatting inconsistencies
- Passenger Count showed duplicate and extreme values
- Overall_Rating contained an invalid non-numeric value ('n')

Profiling Summary Table:

All six attributes had 0% nulls but duplicates were present in every column. No attribute qualified as a key candidate, confirming the need for a separate identifier

Schema Matching

Schema Matching is the process of defining how data from source systems is transferred and transformed into the target data warehouse structure

It ensures that data from different sources can be integrated consistently and accurately

Source A	Source B	Match Type	Confidence	Justification
Aircraft_Type	Flight_Date	No Match	Low	Different semantics; technical model vs temporal date.
Seat_Type	Traveler_Type	No Match	Low	Different categories; cabin class vs passenger type.
Year	Activity Period	Partial Match	Medium	Both represent time, but one includes month and year while the other includes only year
Carrier_Name	Operating Airline	Match	High	Both represent airline company name
airport	GEO Region	Partial Match	Medium	Both represent location, but one is specific and the other is general

Key Finding:

- One direct match was identified between Carrier_Name and Operating Airline
- Two partial matches were found between Year & Activity Period and Airport & GEO Region
- Two attribute pairs were classified as No Match due to differences in meaning and data context
- The matching process improved consistency before moving to the schema mapping stage

Schma Mapping

Schema Mapping defines how matched attributes are transferred and transformed from source systems into the target data warehouse

Required Deliverables

Source Attribute(s)	Target Attribute(s)	Mapping Type	Short Description
Carrier_Name	Operating Airline	1:1	Both represent the airline company name
Year	Activity Period	1:N	One year corresponds to multiple months within the period
Airport	GEO Region	N:1	Multiple airports belong to one geographic region

Three mapping rules

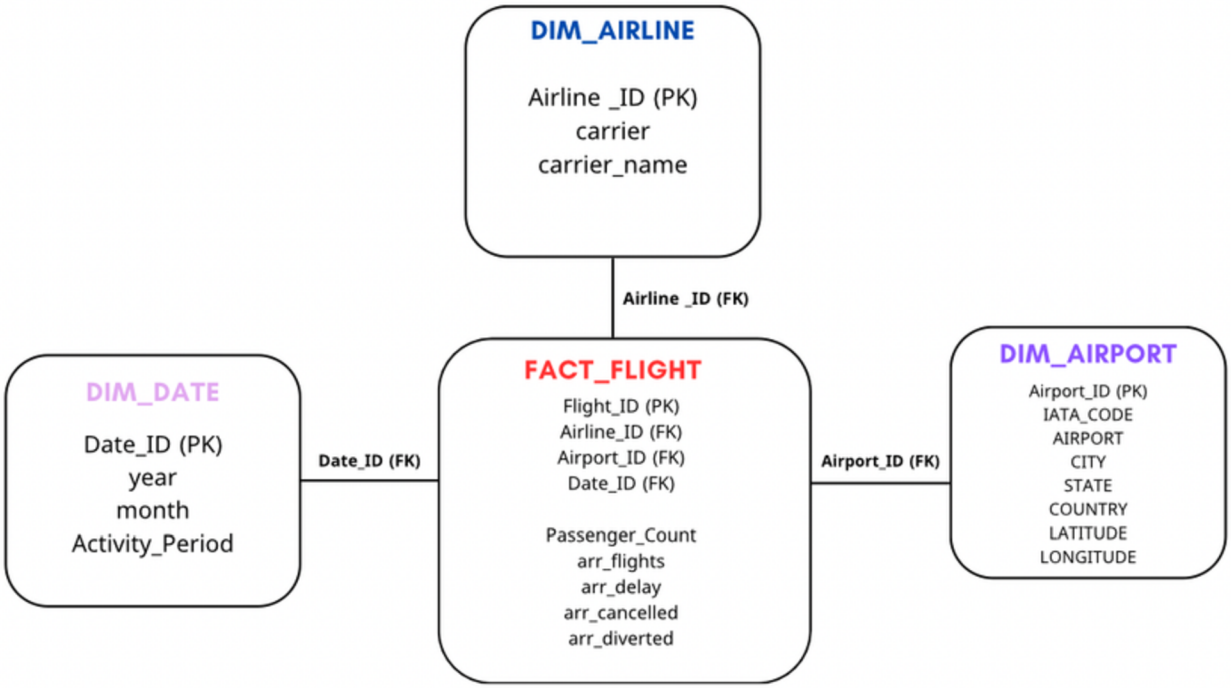
Rule	Rule Expression	Explanation
Rule 1	(SQL-style) SELECT Carrier_Name AS Operating_Airline FROM Source_Table;	The airline name is directly mapped from source to target without any transformation.
Rule 2	(Logical): Year → Activity Period (Year + Month)	A single year is expanded into multiple months within the activity period.
Rule 3	(Logical): Origin Airport → GEO Region	Each airport is mapped to its corresponding geographic region.

Key Finding:

- Three mapping rules were created to transform source data into the warehouse structure
- Different mapping types (1:1, 1:N, and N:1) were used based on data relationships
- The mapping process ensured consistency before ETL implementation

Star Schema & Data Warehous & ETL

A Star Schema was designed to organize airline data into one fact table and multiple dimension tables for efficient analysis and reporting



Extract

Data was collected from multiple airline-related datasets:

Flight Delay Data
Air Traffic Passenger Trends
Flight Delay and Cancellation

Transform

During this stage, the data was prepared for integration by:

Performing Data Cleaning
Applying Schema Matching
Applying Schema Mapping
Selecting relevant attributes
Organizing the data according to the Star Schema structure

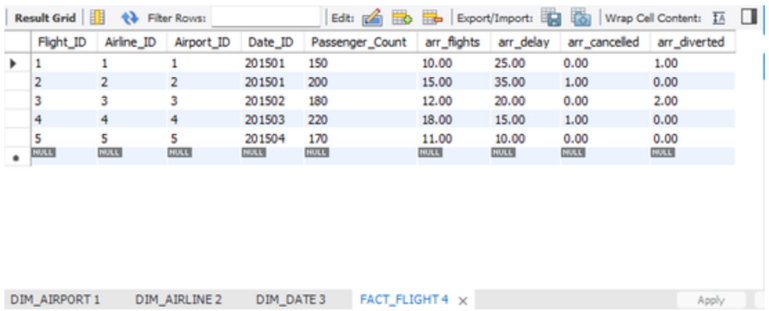
Examples:
Carrier_Name → Operating Airline

Load

The transformed data was loaded into the data warehouse tables:

FACT_FLIGHT
DIM_AIRLINE
DIM_AIRPORT
DIM_DATE

SQL Implementation:

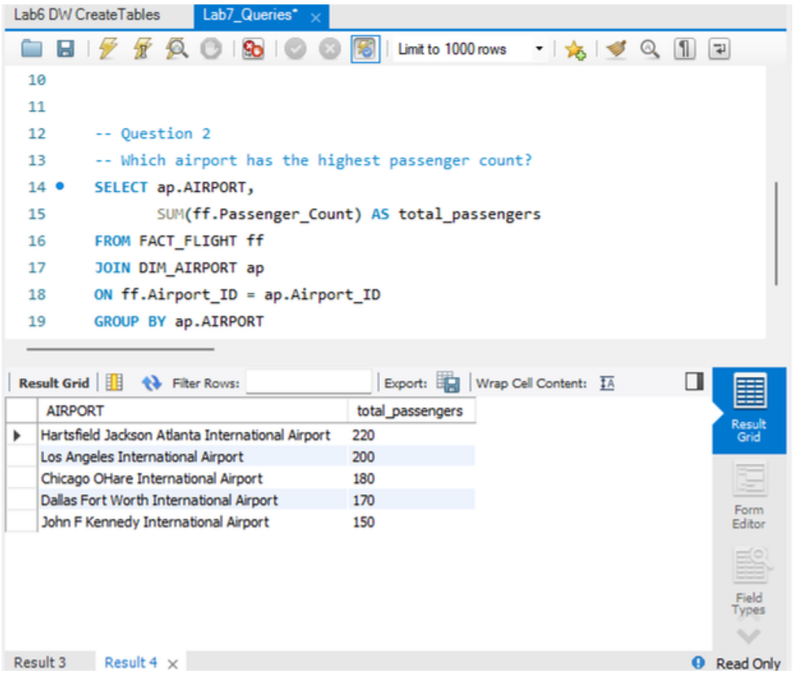


Strategic Question & OLAP/Data Minjng

SQL queries were used to analyze flight warehouse data and answer key business questions:

Business Question	Key Finding
Which airline has the highest number of flights?	Southwest Airlines
Which airport has the highest passenger count?	Hartsfield Jackson Atlanta Airport
Which airline has the highest average delay?	Delta Air Lines
Which month has the highest passenger activity?	Month 1

Example of SQL query used for analysis:



Key Insights & Visual Dashboard



Key Insights

- Southwest Airlines recorded the highest number of flights among all analyzed airlines
- Hartsfield Jackson Atlanta Airport achieved the highest passenger traffic, indicating strong airport activity
- Delta Air Lines showed the highest average delay and may require operational improvements
- Passenger activity peaked during Month 1, suggesting a higher demand period for air travel

Business Value

- Supports data-driven decision making
- Helps identify high-demand periods
- Assists in monitoring airline performance
- Provides insights for resource planning and operational improvements

Recommendations & Lessons Learned

Recommendations

- Increase resources during periods of high passenger activity
- Monitor airlines with high delays to improve operational performance
- Continue updating the data warehouse with new flight data
- Use dashboard analytics to support data-driven decision making

Lessons Learned

- Data cleaning is essential for improving data quality
- Schema matching and schema mapping are important for successful data integration
- ETL processes help organize and load data efficiently into the warehouse
- Dashboards and visualizations make data easier to analyze and interpret
- Star Schema design improves analytical performance and reporting efficiency